MILITARY SPECIFICATION

PROJECTILE, 20MM, TARGET PRACTICE, M55A4B1

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the projectile metal parts for Projectile, 20MM, Target Practice, M55A4B1 manufactured from bar stock by impact extrusion with steel nose cap.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications for

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-651	-	Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

7258861	-	Carton, for Interplant Shipment of Projectiles, 20MM Assemblies and Metal Parts Assemblies (200)
7259136	-	Shipping Container for Inert Projectile, 20MM Assemblies and Metal Parts Assemblies (10,000)
7259191	-	Cartridge, 20MM, for M55 Type, Target Practice, Projectile Metal Parts Test
9342861	-	Projectile, 20MM, Target Practice M55A4B1

INSPECTION EQUIPMENT DRAWINGS

LI 9342861	-	Index of Inspection Equipment Lists for Projectile, 20MM, TP, M55A4B1

PUBLICATIONS

AMCR 715-505	-	Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIELS (ASTM)

ASTM A108	-	Steel Bars, Carbon, Cold-Finished, Standard Quality
ASTM B117	-	Method of Test for Salt Spray (Fog) Testing
ASTM A576	-	Steel Bars, Carbon, Hot-Rolled, Special Quality

(Application for copies should be addressed to the American Society for Testing and Materiels, 1916 Race Street, Philadelphia, PA 19103)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The projectile shall be in accordance with the requirements of Drawing (Dwg.) 9342861 and with all requirements of the drawings and specifications shown thereon.

3.2 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without prior approval. (See 6.3)

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Banding. The rotating band blank and the rotating band seat shall be clean and dry at the time of banding. Shot or sandblasting shall not be used to clean the band seat after knurling. The rotating band blank shall fill the band seat.

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3.5 Soundness. Metal discontinuity in the projectile body, as determined by eddy current inspection, shall be less than that represented by Production Check Standard (PCS) No. 1 (see 6.5).

3.6 Protective coating.

3.6.1 Phosphate coating. The pretreatment cleaning process used shall be in accordance with Dwg. 9342861. The phosphate coating shall comply with the requirements of TT-C-490 for appearance and minimum weight, as applicable to the type of phosphate coating used. Test specimens (panels) shall be in accordance with the requirements of TT-C-490.

3.6.2 Paint coating. The paint of the projectile metal parts shall be in accordance with applicable requirements of Dwg. 9342861. Exterior paint adhesion shall be in accordance with applicable requirements of TT-C-490.

3.6.3 Corrosion resistance. The painted surfaces of the projectile shall show no evidence of corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 24 hours.

3.7 Metal parts security. The sample projectiles, assembled into test cartridges loaded in accordance with the requirements of Dwg. 7259191, shall not separate or breakup in the test for metal parts security.

3.8 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Processing defects. The projectile shall be free of defects such as, dents, deep scratches, cuts, sharp edges, burrs, metal flaws, excessive surface roughness and other forms of substandard workmanship.

3.8.2 Cleanliness. The projectile shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. After inspection by the contractor and provisional acceptance by the Government, the contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as indicated in Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078. Failure of the original first article sample to be in accordance with applicable drawings and the requirments of Para. 3, as specified herein, will result in sample disapproval. (See 6.2).

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63438(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Projectile and Components				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rotating Band Blank Dwg. 7258816	32		3.1	4.4.2.1
	Nose, Projectile Dwg. 9342858	125		3.1	4.4.2.2
	Projectile Body Dwg. 9342859	125		3.1	4.4.2.3
	Projectile (Unpainted) Dwg. 9342860)	125		3.1	4.4.2.4
	Soundness Verification	125	0-1 *	3.5	4.5.4
	Projectile MPTS (Coated prior to painting) (Dwg 9342861)				
	Standard Test Panels (specimens)	3	0-1 *	3.6.1	4.5.5
	Projectile MPTS (Painted) (Dwg. 9342861)	125		3.1	4.4.2.5
	Band Tightness	125	0-1 *	3.4	4.5.1
	Paint Adhesion	13	0-1 *	3.6.2	4.5.6
	Corrosion Resistance	32	0-1 *	3.6.3	4.5.7
	Metal Parts Security Test	4000	0-1 *	3.7	4.5.8

NOTES:

* Accept - Reject Criteria

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4.4 Quality conformance inspection.

4.4.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.4.1.1 Lot. A lot shall consist of projectile metal parts produced by one manufacturer in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision. (See 3.2).

4.4.2 Examination. See MIL-A-48078.

a. MIL-STD-651 shall apply in defining and evaluating visual defects.

b. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63438(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7258816
4.4.2.1	Rotating Band Blank				NEXT HIGHER ASSEMBLY 9342860
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Foreign matter		.25%	3.8.2	Visual
102.	Defective metal, including lamination		.25%	3.5	Visual
103.	Incomplete or deformed		.25%	3.5	Visual 1/
MINOR:					
201.	Width, max.		.40%	3.1	Gage
202.	Evidence of poor workmanship		.65%	3.8	Visual

NOTES: 1/ A rotating band blank which is deformed or is judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions shall be classed defective.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63438(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Nose, Projectile	1 OF 1	9342858
			NEXT HIGHER ASSEMBLY 9342860

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Thickness, datum area		.25%	3.1	Gage
102.	Depth, to spherical surface		.25%	3.1	Gage
103.	Length, crimping lip, min.		.25%	3.1	Gage
104.	Foreign matter		.25%	3.1	Visual
MINOR:					
201.	Length, crimping lip, max.		.40%	3.1	Gage
202.	Angle, crimping lip with conical surface		.40%	3.1	Gage
203.	Inside diameter, crimping lip		.40%	3.1	Gage
204.	Profile, closed end		.40%	3.1	Gage
205.	Thickness, except datum area		.40%	3.1	Gage
206.	Evidence of poor workmanship		.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63438(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9342859
4.4.2.3	Projectile Body				NEXT HIGHER ASSEMBLY 9342860
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Knurling missing or incomplete		.25%	3.1	Visual 1/
102.	Diameter, bourrelet, min.		.25%	3.1	Gage
103.	Length, overall, max.		.25%	3.1	Gage
104.	Length, mouth to rear edge of rotating band seat		.25%	3.1	Gage
105.	Depth, knurling, min.		.25%	3.1	Visual/Gage 1/
106.	Length, rear edge of rotating band seat to rear edge of pinch groove		.25%	3.1	Gage
107.	Diameters (upper & lower), pinch groove		.25%	3.1	Gage
108.	Width of pinch groove		.25%	3.1	Gage
109.	Radius, pinch groove, max		.25%	3.1	Gage
110.	Foreign matter in rotating band seat		.25%	3.1	Visual (See 3.4)
MINOR					
201.	Diameter, rotating band seat (knurled)		.40%	3.1	Gage
202.	Width, rotating band seat		.40%	3.1	Gage
203.	Inside diameter, cavity		.40%	3.1	Gage
204.	Width, knurling, min		.40%	3.1	Visual/Gage 1/
205.	Thickness, outside wall at mouth		.40%	3.1	Gage
206.	Evidence of poor workmanship		.65%	3.8	Visual

NOTES: 1/ Measurement of knurling for compliance with the minimum depth and width requirements shall apply where visual inspection indicates that the depth or width dimension is borderline.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63438(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.4	Projectile, Uncoated		9342860
			NEXT HIGHER ASSEMBLY
			9342861

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:					
1.	Soundness (Ferrous metal discontinuity test)		100%	3.5	See 4.5.2
2.	Crack or split in nose		100%	3.1	Visual
MAJOR:					
101.	Runout, nose with datum surface		.25%	3.1	Gage
102.	Diameter, bourrelet, min.		.25%	3.1	Gage
103.	Loose nose		.25%	3.1	Manual 1/
104.	Body not flush or below at junction with nose		.25%	3.1	Visual
105.	Length, overall		.25%	3.1	Gage
106.	Length, nose to rear edge of rotating band		.25%	3.1	Gage
107.	Diameter, body, rear of rotating band, min.		.25%	3.1	Gage
108.	Diameter, rotating band		.25%	3.1	Gage
109.	Runout, rotating band with datum surface		.25%	3.1	Gage

NOTES: 1/ If the nose moves with respect to the projectile body under thumb and finger pressure applied in the laterial or axial direction, the projectile shall be classified defective.

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MIL-P-63438(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER 9342860
4.4.2.4	Projectile, Uncoated		NEXT HIGHER ASSEMBLY 9342861

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
110.	Runout, body, rear of rotating band with datum surface		.25%	3.1	Gage
111.	Location, rear crimp groove		.25%	3.1	Gage
112.	Profile, rear crimp groove		.25%	3.1	Gage
113.	Void at junction of rotating band and projectile body		.25%	3.4	Visual
114.	Radius, rear edge of crimp groove, max		.25%	3.1	Gage
115.	Radius, base of projectile		.25%	3.1	Visual/Gage 2/
MINOR:					
201.	Finish improper		.40%	3.6.1	Visual 1/
202.	Diameter of bourrelet, max.		.40%	3.1	Gage
203.	Depth and width of undercut, front and rear of rotating band, max.		.40%	3.1	Gage
204.	Width of rotating band		.40%	3.1	Gage
205.	Evidence of poor workmanship		.65%	3.8	Visual

NOTES: 1/ Measurement of the surface roughness shall apply where visual inspection indicates that the specified finish is exceed. 2/ Measurement of the base radius shall apply where visual inspection indicates that the base radius of the projectile is borderline.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9342861
4.4.2.5	Projectile, (Painted)		NEXT HIGHER ASSEMBLY 7258817

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Diameter, bourrelet, max.		.25%	3.1	Gage
102.	Diameter, body, rear of rotating band, max.		.25%	3.1	Gage
103.	Weight		.25%	3.1	Balance
MINOR:					
201.	Marking incorrect, incomplete, illegible, or missing		.40%	3.1	Visual
202.	Paint on rotating band		.40%	3.6.2	Visual
203.	Rotating band damaged		.40%	3.1	Visual
204.	Evidence of poor workmanship		.65%	3.8	Visual

NOTES:

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4.4.3 Testing. The tests listed in Table II shall be performed on each lot in accordance with the test methods prescribed in 4.5. The samples shall be selected in such a manner as to be representative of the entire lot. The sample size and acceptance criteria for each test shall be as specified.

TABLE II. Testing.

Test	Sample Size		Item	Rqmt Para
Band tightness	1/	315	Projectiles, painted (or uncoated) (See 4.5.1)	3.4
Automatic inspection effectivenss	2/	--	PCS No. 1 (See 4.5.3)	--
Soundness verification	3/	315	Projectiles, uncoated (See 4.5.4)	3.5
Phosphate coating (appearance & weight)	4/	3	Projectiles, (before painting) samples or standard panels (See 4.5.5)	3.6.1
Paint adhesion	5/	13	Projectiles, painted (See 4.5.6)	3.6.2
Corrosion resistance	6/	32	Projectiles, painted (See 4.5.7)	3.6.3
Metal parts security	7/	250	Projectiles, painted (See 4.5.8)	3.7

1/ If two or more units of the sample have an average radial void, between the seated band and the band seat, greater than 0.004 inch, the lot shall be rejected. If one unit of the sample has an average radial void greater than 0.004 inch, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units have an average radial void greater than 0.004 inch, the lot shall be rejected.

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2/ Failure of the flaw detector to reject PCS No. 1 shall be cause for rejection of all assemblies inspected for "soundness" since the preceding test and shall require requalification of the equipment for automatic inspection (See 4.5.3).

3/ Rejection of one or more units of the sample by the flaw detector in the test for body metal soundness shall be cause for rejection of the lot.

4/ Failure of the phosphate coating on the sample projectiles or standard panels to comply with the applicable requirements shall be cause for rejection of the lot (See 3.6.1).

5/ If two or more units of the sample fail to comply with the requirement for paint adhesion, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the requirement for paint adhesion, the lot shall be rejected.

6/ If two or more units of the sample fail to comply with the corrosion resistance requirement, the lot shall be rejected. If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirement, the lot shall be rejected.

7/ The lot will be rejected, if any one of the following types of metal parts separation or breakup occurs in the metal parts security test, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s).

a. Body breakup.

b. Loss of nose.

c. Complete rotating band separation. This is defined as detachment of all of the band having band seat knurling imprint.

d. Partial rotating band separation. This is defined as detachment of part of the band having band seat knurling imprint.

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e. The following acceptance criteria shall apply to thrown band fragments having no evidence of band seat knurling imprint:

(1) The lot will be accepted if two or less projectile assemblies throw band fragments having length not greater than 0.500 inch.

(2) The lot will be rejected if three or more projectile assemblies throw band fragments having length not greater than 0.500 inch.

(3) The lot will be rejected if any projectile assembly throws a band fragment having length greater than 0.500 inch (indicative of deep-seated band lamination).

(4) The lot will not be penalized for band particles thrown as a result of normal band fringing or slivering.

4.4.4 Packaging, packing and marking inspection. Inspection for packaging, packing and marking shall comply with applicable requirements of MIL-STD-1169.

4.4.5 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method Column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

4.5 Test methods and procedures.

4.5.1 Band tightness. The method of test shall comply with all requirements of the operating instructions for the band tightness tester (Dwg. 7259545 a detail of LI 9342861). The inspection sample shall be cycled through the tester to determine compliance with the rotating band tightness requirement. The total measurement of radial void, registered on the tester indicator at final ram pressure, shall be divided by four (number of indentors) to determine the average radial void.

4.5.2 Soundness. The method of inspection shall comply with the requirements of Operating Instructions for Eddy Current Flaw Detector (Dwg. D11075246, a detail of LI 9342861), except for those inspection requirements which are reserved for Government application, as specified herein.

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4.5.2.1 Application of defect standards. Master defect standards, referenced in the operation instructions, shall apply to resolve inspection discrepancies occurring between the results of defect standards supplied for contractor "soundness" inspection and those supplied for Government "soundness" verification. (See 6.5)

4.5.3 Automatic eddy current inspection effectiveness. Equipment checkout shall be performed as stated in the operating instructions at the start of each shift and after any adjustment or shutdown. Upon completion of checkout, cycle the PCS No. 1, front slot and rear slot standards through the inspection system during automatic operation. All three standards must be rejected to assure correct adjustment for reject level and scan limits. During remainder of shift the PCS No. 1 shall be cycled through the equipment at intervals of 20 $\pm$ 5 minutes during automatic operation.

4.5.3.1 Equipment verification. The government inspector will maintain, independent of the contractor, a set of standards and may cycle these standards through the equipment at any time. Failure of equipment to reject standards will be evidence that the equipment is not properly adjusted and the contractor will be so notified.

4.5.4 Soundness verification. After equipment checkout and verification as specified in 4.5.3 and 4.5.3.1 the entire lot sample will be inspected with the equipment operating automatically. Any rejected unit of the sample will be automatically cycled through the equipment with the CAL. button pressed (to lower the sensitivity one step below that used for production inspection). Any unit which is rejected at the lower sensitivity shall be classed as a defective.

4.5.5 Phosphate coating. The method of test shall be as prescribed in TT-C-490 for the type of coating used (see Dwg. 9342861). The prescribed sample projectiles or standard panels shall be processed at the start of daily production and at 4-hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

4.5.6 Paint adhesion. The method of test shall be as prescribed in TT-C-490. The test shall be made on the exterior painted surface of each projectile of the sample forward of the rotating band. Visible exposure of underlying phosphate coating or ferrous metal of any unit of the sample, as a result of the test, shall be recorded as a defect.

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4.5.7 Corrosion resistance. The method of test shall be as prescribed in American Society for Testing and Materiels (ASTM), ASTM B117. The paint shall not be disturbed, however, the rotating band and unpainted surfaces of the sample projectiles may be masked out by a wax coating or other suitable mateial. The salt spray solution and exposure time shall be as specified in 3.6.3.

4.5.8 Metal parts security. The method of test will be as prescribed in AMCR 715-505, Volume 8, as applicable. The projectiles will be assembled as specified on Dwg. D7259191, Cartridge, 20MM, for M55 Type, Target Practice, Projectile Metal Parts Test. The test cartridges, at ambient temperature, will be fired in bursts of 50 shots in an M39 gun. The gun barrel will be at ambient temperature at the beginning of each test and cooled to ambient temperature after each burst. The cyclic rate of fire will be recorded for informational purposes.

5. PACKAGING

5.1 Packing, Level C. Each lot of product shall be packed as specified by Dwg. 7258861 or Dwg. 7259136, as directed by the procuring activity.

5.2 Marking and labeling. Marking or labeling of packed cartons shall comply with applicable requirements of MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify:

a. Title, number and date of this specification.

b. Place of inspection, if not at place of manufacture.

c. Packing and marking instructions. (See 5.1 and 5.2)

d. First article sample requirements. (See 3.3, 4.3 and 6.2)

e. Provision for shipment and testing of first article and production lot samples for metal parts security test. (See 4.3 and 4.5.8)

f. Provision for the supply, maintenance and disposition of Government furnished inspection equipment for acceptance inspection purposes.

g. Provision for the submission and approval of the manufacturing process. (See 3.2)

h. Provisions for the inclusion of MIL-STD-1167, Ammunition Data Cards on DD 1423, Contract Data Requirements List.

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6.2 First article sample. Where a first article sample is required by the contract, the contracting officer will notify the test facility when to proceed with the metal parts security test. Additional assemblies may be required by the test facility. (See 4.3)

6.3 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

6.4 Intermediate point inspection. The classification of defects identifies the defect characteristics (among other things) for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality of the end item.

6.5 Defect standards. The initial supply of defect standards and instructions for contractor and Government application to the eddy current method of inspection, will be furnished with the inspection equipment. Replacement standards shall be requisitioned through normal channels by the Government agency having contract administration responsibility.

Custodian:
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